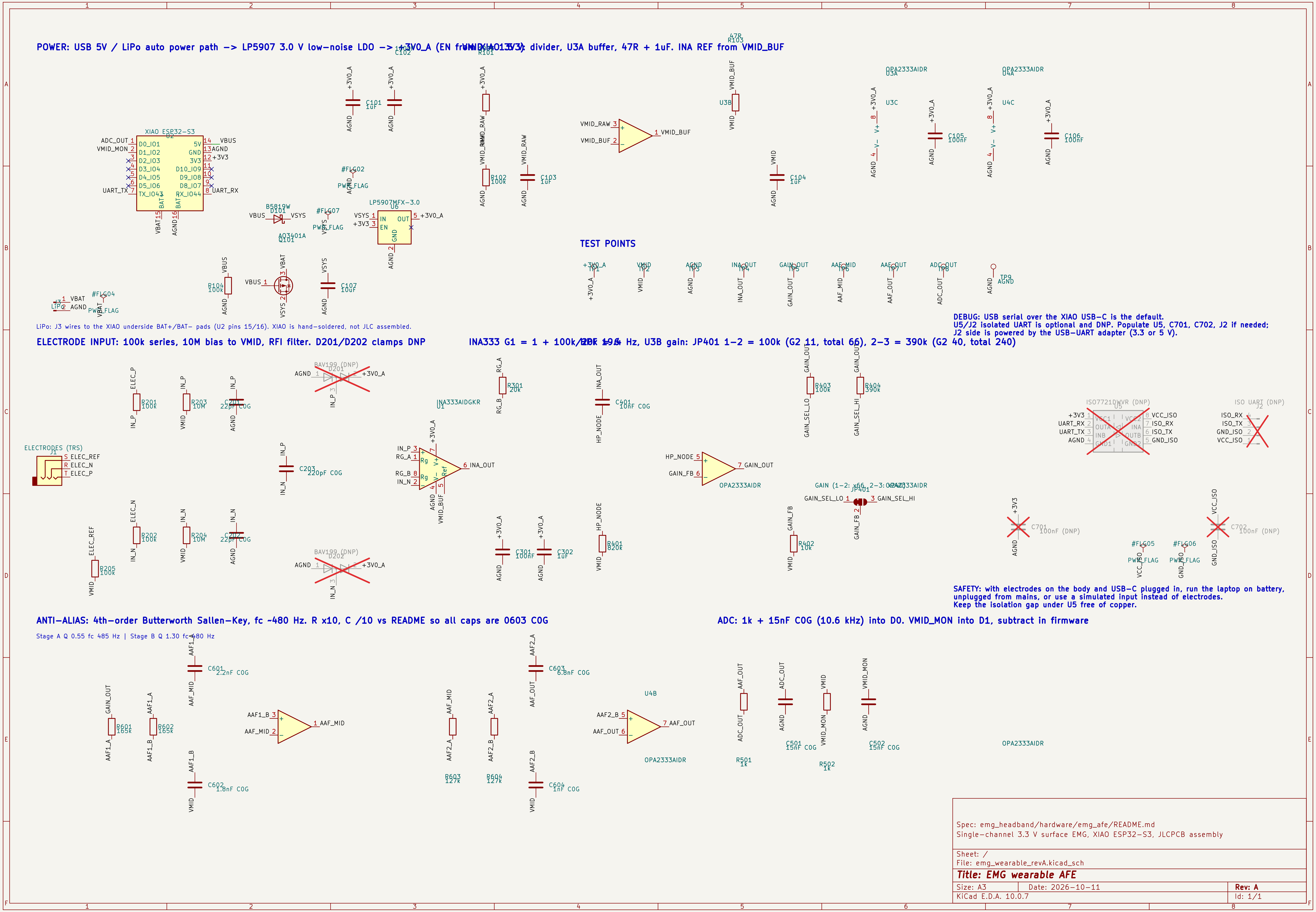

POWER: USB 5V / LiPo auto power path -> LP5907 3.0 V low-noise LDO -> +3V0_A (EN from XIAO 3V3)

VMID 1.5 V: divider, U3A buffer, 47R + 1uF. INA REF from VMID_BUF

LiPo: J3 wires to the XIAO underside BAT+/BAT- pads (U2 pins 15/16). XIAO is hand-soldered, not JLC assembled.

TEST POINTS

DEBUG: USB serial over the XIAO USB-C is the default.
U5/J2 isolated UART is optional and DNP. Populate U5, C701, C702, J2 if needed;
J2 side is powered by the USB-UART adapter (3.3 or 5 V).

ELECTRODE INPUT: 100k series, 10M bias to VMID, RFI filter. D201/D202 clamps DNP

INA333 G1 = 1 + 100k/20k = 6

HPF 19.4 Hz, U3B gain: JP401 1-2 = 100k (G2 11, total 66), 2-3 = 390k (G2 40, total 240)

SAFETY: with electrodes on the body and USB-C plugged in, run the laptop on battery,
unplugged from mains, or use a simulated input instead of electrodes.
Keep the isolation gap under U5 free of copper.

ANTI-ALIAS: 4th-order Butterworth Sallen-Key, fc ~480 Hz. R x10, C /10 vs README so all caps are 0603 C0G

Stage A Q 0.55 fc 485 Hz | Stage B Q 1.30 fc 480 Hz

ADC: 1k + 15nF C0G (10.6 kHz) into D0. VMID_MON into D1, subtract in firmware

Spec: emg_headband/hardware/emg_afe/README.md
Single-channel 3.3 V surface EMG, XIAO ESP32-S3, JLCPCB assembly

Sheet: /
File: emg_wearable_revA.kicad_sch

Title: EMG wearable AFE

Size: A3 Date: 2026-10-11 Rev: A
KiCad E.D.A. 10.0.7 Id: 1/1